

TOOLS

Applied Ecosystem	GRADUATED				Julia Notebooks					
	GRADUATED									
Engineering & Simulation	Simulation & Modeling					FEA & Structural Analysis				
	GRADUATED	GRADUATED	GRADUATED	GRADUATED		GRADUATED	GRADUATED	GRADUATED		
CAD & Design	GRADUATED	GRADUATED	GRADUATED				Infrastructure Design			
	GRADUATED			GRADUATED						
Chemistry & Molecular Modeling	Quantum Chemistry					Chemistry Tools				
	GRADUATED	GRADUATED	GRADUATED	GRADUATED		GRADUATED	GRADUATED	GRADUATED		
	Computational Chemistry					Molecular Visualization				
	GRADUATED	GRADUATED	GRADUATED	GRADUATED		GRADUATED	GRADUATED			
Bioinformatics	Language-Specific Tools					Statistical Platforms				
	GRADUATED	GRADUATED				GRADUATED				
	Python Tools									
	GRADUATED									
Data Science & Programming	Machine Learning Frameworks					Scientific Computing				
	GRADUATED	GRADUATED	GRADUATED	GRADUATED	GRADUATED	GRADUATED	GRADUATED			
	GRADUATED					GRADUATED				
	Specialized ML Tools					Data Visualization				
	GRADUATED	GRADUATED	GRADUATED	GRADUATED	GRADUATED	GRADUATED	GRADUATED	GRADUATED	GRADUATED	
	Numerical & Mathematics					JavaScript ML				
	GRADUATED	GRADUATED	GRADUATED	GRADUATED	GRADUATED	GRADUATED	GRADUATED	GRADUATED		
	Programming Languages					Business Analytics				
	GRADUATED	GRADUATED	GRADUATED	GRADUATED				GRADUATED		
	Big Data & Distributed Computing									
	GRADUATED	GRADUATED	GRADUATED	GRADUATED						
	Location & Mapping Platforms					GIS Platforms				
Geospatial & Mapping	GRADUATED	GRADUATED	GRADUATED	GRADUATED	GRADUATED	GRADUATED	GRADUATED	GRADUATED		
	Web Mapping Libraries					3D & Advanced Visualization				
	GRADUATED	GRADUATED	GRADUATED				GRADUATED	GRADUATED		
	Geospatial Data & Tools					Earth Observation				
	GRADUATED	GRADUATED	GRADUATED				GRADUATED			
	DFT & Electronic Structure					Educational & Reference				
Physics & Material Science	GRADUATED	GRADUATED	GRADUATED	GRADUATED	GRADUATED	GRADUATED	GRADUATED			
	GRADUATED	GRADUATED	GRADUATED	GRADUATED	GRADUATED					
	Nuclear & Particle Physics					Molecular Dynamics				
	GRADUATED	GRADUATED	GRADUATED	GRADUATED	GRADUATED	GRADUATED				
Space Science & Astronomy	NASA Platforms & Programs					Protocol & Observation Planning				
	GRADUATED	GRADUATED	GRADUATED	GRADUATED				GRADUATED	GRADUATED	
	Data & Modeling Centers					Space Science References				
	GRADUATED	GRADUATED	GRADUATED	GRADUATED				GRADUATED		
	Jupyter Tools									
	GRADUATED	GRADUATED								
Neuroimaging & Brain Computing	Neuroimaging Platforms					Brain-Computer Interface				
	GRADUATED	GRADUATED	GRADUATED	GRADUATED		GRADUATED	GRADUATED			
	Comprehensive Platforms					Reference & Resources				
	GRADUATED	GRADUATED	GRADUATED				GRADUATED			
Data & Visualization Resources	Data Catalogs & APIs					Databases				
	GRADUATED	GRADUATED				GRADUATED				
	Visualization & Algorithm Tools									
	GRADUATED									
Languages & Frameworks	Architecture & Deployment					Build Tools				
	GRADUATED	GRADUATED	GRADUATED	GRADUATED	GRADUATED	GRADUATED	GRADUATED			

The diagram illustrates the Cloud Native Computing Foundation (CNCF) landscape, organized into a grid with rows representing different categories and columns representing different layers of the cloud-native stack. Each cell contains icons representing various projects, many of which are marked as 'GRADUATED'.

Rows (Categories):

- NET:** Frameworks & Initiator, Framework & Runtime, Build & Package Management.
- Platform:** Web Frameworks, Application Servers, Package Management.
- Data:** Web & Async Frameworks, Package Management & Build, Language & Toolchain.
- Cloud:** Build Generators, Build System.
- Ops:** Tools & Middleware, Simulation & Modeling, Core Framework.
- Ops:** Language & Development, Simulators & Hardware Providers.
- Java:** Scientific Computing, Language & Runtime, Machine Learning & Data Science, Package Management, Visualization.

Columns (Layers):

- Application:** Frameworks & Initiator, Framework & Runtime, Build & Package Management, Web Frameworks, Application Servers, Package Management.
- Infrastructure:** Web & Async Frameworks, Package Management & Build, Language & Toolchain.
- Cloud:** Build Generators, Build System.
- Ops:** Tools & Middleware, Simulation & Modeling, Core Framework.
- Ops:** Language & Development, Simulators & Hardware Providers.
- Java:** Scientific Computing, Language & Runtime, Machine Learning & Data Science, Package Management, Visualization.

Legend:

- GRADUATED:** Projects marked with a green leaf icon.
- Incubating:** Projects marked with a grey square icon.

Compute Services

The diagram illustrates the architecture of various cloud providers, categorized into Google Cloud, Microsoft Azure, Oracle Cloud Infrastructure, IBM Cloud, AWS, and OpenStack. Each provider's architecture is represented by a grid of service icons, with some icons highlighted in blue to indicate specific services or components. The diagram is organized into rows and columns, with each row representing a different cloud provider and each column representing a different service category. The services are grouped into functional areas, such as Compute, Storage, Networking, and Analytics, and are represented by icons that are either blue (active) or grey (inactive). The diagram is a complex visual representation of cloud architecture, showing the relationships between different services and how they are organized within each provider's ecosystem.

Google Cloud

- AI/ML Services
- Database Services
- Networking & Content Delivery
- OCI Applications
- Analytics Services
- Storage Services
- Compute Services
- Data Integration & Migration

Microsoft Azure

- Storage Services
- Data Integration & Migration
- Networking & Content Delivery
- Storage Services
- Compute Services
- Analytics Services
- AI/ML Services
- Azure Applications
- Database Services
- Data Integration & Migration

Oracle Cloud Infrastructure

- OCI Applications
- Networking & Content Delivery
- Compute Services
- Data Integration & Migration

IBM Cloud

- OCI Applications
- Networking & Content Delivery
- Compute Services
- Analytics Services
- Storage Services
- Database Services
- Data Integration & Migration

AWS

- OCI Applications
- Networking & Content Delivery
- Compute Services
- Analytics Services
- Storage Services
- Database Services
- Data Integration & Migration

OpenStack

- OCI Applications
- Networking & Content Delivery
- Compute Services
- Analytics Services
- Storage Services
- Database Services
- Data Integration & Migration